

Kai Wu

+86 132****2782 | 22125161@zju.edu.cn | 55kwukai.github.io

EDUCATION

Zhejiang University

M.Eng. in Mechanical Engineering

- Filed 9 Chinese invention patent applications; 5 granted.

Hangzhou, China

Sep. 2021 - Mar. 2024

Dalian Maritime University

B.Eng. in Rescue and Salvage Engineering

- National First Prize, 12th Chinese Mathematics Competition for College Students.

Dalian, China

Sep. 2017 - Jun. 2021

PROFESSIONAL EXPERIENCE

Manycore Tech | AI Product Manager

Jul. 2024 - Jun. 2026

Domain-Specific AI Coding Agent for 3D Modeling | [Hugging Face](#)

- Led product roadmap planning for a Text-to-Script-to-3D system, enabling high-precision parametric modeling scripts to be generated from natural-language descriptions of complex geometry.
- Designed and coordinated implementation of a ReAct-based agent loop spanning requirement interpretation, code generation, script execution, scene inspection, output verification and iterative correction.
- Designed a three-layer verification and feedback mechanism (execution, semantic and rule-based) to detect and correct geometric errors even when generated code executed successfully.
- Established a benchmark for spatial modeling with metrics for code executability, geometric similarity and constraint consistency; used a representative test dataset to track agent performance across iterations.

3D Model Quality Inspection Platform | Web IDE and DSL

- Built an agent harness to run automated rule checks after 3D generation and reject invalid models, addressing gaps in agent verification.
- Designed a domain-specific language (DSL) to describe model features and automate quality checks based on recurring issues in real-world use.
- Delivered test case management, batch validation and result tracking for a platform with 533 DSL scripts and over 10 million model analyses.

AI Customer Support for Kujiale | 3D Spatial Design Software

- Designed LLM- and RAG-based customer support workflows with conversational state management for parameter adjustments, manufacturing-process queries, asset retrieval and model inspection; optimized multi-turn interactions.
- Designed intent-based routing and introduced hybrid search and reranking, improving retrieval accuracy for complex queries by 27%.
- Standardized structured model outputs and defined fallback handling for knowledge-base misses and human handoff, improving client-side response handling and the support experience.

PROJECTS

SimReady CLI | [GitHub](#)

Dec. 2025 - Jan. 2026

- Built a tool for agent-driven CAD simulation workflows that converts hierarchical parametric models into physics-ready MJCF assets, enabling MuJoCo motion simulation after adding joint constraints.
- Implemented model hierarchy parsing, joint-constraint mapping, rigid-body hierarchy reconstruction, joint and actuator generation, and mesh processing to preserve structure, poses and motion limits in simulation.
- Converted static CAD models into articulated, interactive simulation assets to support object understanding, action generation and environmental feedback in world models.

On-Device AI Keyboard Prototype | Text-to-Emoji Fine-Tuning | [Hugging Face](#)

Oct. 2025 - Nov. 2025

- Designed an on-device AI keyboard prototype using an LLM to convert context-rich text into emojis for young users' social messaging and emotional expression.
- Selected Gemma-3-270M-it for on-device privacy and latency requirements; applied task-specific QLoRA fine-tuning to improve output consistency.
- Completed post-training, evaluation and Hugging Face deployment; assessed on-device inference cost and latency, and verified that fine-tuning outperformed a prompt-only baseline.

SKILLS

- Tools: C#, Python, HTML, SQL, LaTeX, Git, Gurobi, Postman; [prototype development](#) and technical design review.
- AI Product Development: Agent Loop, KV Cache, Tool Use, MCP, Memory, Subagent, Fallback, Verification, Evaluation, Post-Training.
- English: CET-6.